Ruyi Tao

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School of Systems Science, Beijing Normal University
Research Interests: Multi-Scale Complex System Modeling, Scaling Theory, Growth Dynamics, Innovation
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Experience

Beijing Institute of Mathematical Sciences and Applications: Visiting Scholar, 2026.5-present

Sante Fe Institute: Visiting Scholar, 2024.5, 2024.8

Beijing Normal University, Systems Science: Ph.D. Candidate., 2021-present

Hangzhou Dianzi University,Digital Media Technology: Bachelor., 2013-2017

Popular Science Articles

- **The Trade-off Between Efficiency and Risk: Understanding the Economic World at the Edge of Criticality.**
- **Two Explanations for Power Laws: Simon vs. Mandelbrot**
- **Economics Meets Physics: An Introduction of Granular Model for Company Growth.**
- **Renormalization Group And Machine Learning: Exploring the Intrinsic Unity of Complex Systems from a Multi-Scale Perspective**.

Activities

- One of the Chinese translator of book *Make Sense of Chaos:A Better Economics for a Better World.* Published in 2026.09
- When Physics Meets Economics: The Growth and Fluctuation of Company Dynamics. *Seminar, 2026.04, Beijing*
- Company Growth Modeling Based on Scaling Theory. *Seminar. 2026.03, Hong Kong*
- Is the Whole Greater Than the Sum of Its Parts? From Complexity Science and Emergence to Consciousness. *Seminar. 2025.12, Beijing*
- Phase Plane: Two-Dimensional Nonlinear Dynamical Systems. *Reading Group. 2025.11, Beijing*
- SFI Classic Papers Reading Group. *Reading Group. 2025.10, Beijing*
- Fractal, Self-Similarity and Power Law. *Reading Group. 2025.9, Beijing*

Publications

- **R Tao,** V. R. Cappelli, W Kai, C. P. Kempes, M. J. Hamilton, G. B. West, J Zhang. **Predicting company growth using scaling theory informed machine learning.** *npj Complex* (2026).

We propose a Scaling-Theory-Informed Machine Learning (STIML) framework that integrates a scaling-based growth model to capture the mechanism-driven average trend, together with a data-driven forecasting model to learn the residual fluctuations. Our model captures both the intrinsic growth mechanisms and the fluctuations, demonstrating superior predictive performance compared with methods that use time series techniques alone.

- **R Tao**, J Zhang ,V. R. Cappelli, C. P. Kempes, M. J. Hamilton, G. B. West. **Scaling laws and a general theory for the growth of public companies.** https://arxiv.org/abs/2109.10379 *(In the process of submission)*

  We develop a scaling framework that focuses on company size as the critical feature determining a variety of tradeoffs, and use this to reveal novel systematic behavior across the diversity of publicly-traded companies. Our model produces predictions that are in good agreement with data for the US, China and Europe market, and we observe that, on average, in both countries sales scale sublinearly with assets and exhibit nearly identical exponents leading, surprisingly and nontrivially, to assets that grow as a power law in time rather than exponentially, as often assumed.

- **R Tao,** N Tao, Y You, J Zhang. **Data driven modeling of self-similar dynamics.** *Physica D: Nonlinear Phenomena,* 2025, 472: 134505.

  We introduce a multiscale neural network framework that incorporates self-similarity as prior knowledge, facilitating the modeling of self-similar dynamical systems.The framework allows us to extract scale-invariant kernels from the dynamics for modeling at any scale and identify the power-law exponents in self-similar systems

- J Zhang , **R Tao**, B Yuan. . **Dynamical reversibility and a new theory of causal emergence.** *npj Complexity*, 2025, 2(1): 3.

- Z Zhang, **R Tao**, Y Tao, J Zhang. (2022). **Graph auto-encoders for network completion.** (Published in *Machine Learning Research*).

Skills

- Python, Pytorch, Matlab, mathematica, JavaScript, SQL